
The Role of Cryotherapy in the Management of Retinoblastoma

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Between February 1974 and August 1987, we treated 67 retinoblastomas in 47 eyes of 45 patients with cryotherapy (triple freeze-thaw technique). Overall, cryotherapy eradicated 53 (79%) of the tumors, whereas additional treatment with episcleral plaque radiotherapy, external beam radiotherapy, or enucleation was necessary in 14 (21%) of the tumors. Tumor destruction was achieved with one or more cryotherapy treatments in all cases in which the tumor was no greater than 2.5 mm in diameter and 1.0 mm in thickness, and in which the tumor was confined to the sensory retina without seeding into the adjacent vitreous humor. We therefore recommend cryotherapy in such cases. Cryotherapy alone failed in 14 larger tumors, ten of which had clinical evidence of vitreous seeding by tumor cells. Cryotherapy is therefore contraindicated in cases of vitreous seeding or if the tumor exceeds 3.5 mm in diameter and 2.0 mm in thickness. Cryotherapy can be successful for tumors up to 3.5 mm in diameter and 2.0 mm in thickness, but more than one treatment may be necessary.

THE THERAPEUTIC OPTIONS in the management of retinoblastoma include enucleation, external beam radiotherapy, episcleral plaque radiotherapy, photocoagulation, and cryotherapy.^{1,3} Cryotherapy has been occasionally used in the treatment of selected patients with retinoblastomas for at least 20 years.^{4,9} Described herein are our technique, current indications,

contraindications, complications, and results of this therapeutic modality based on 13 years of experience with cryotherapy for retinoblastomas at the Ocular Oncology Service at Wills Eye Hospital.

Subjects and Methods

We reviewed the charts of all patients with retinoblastoma who were examined and treated in the Ocular Oncology Service at Wills Eye Hospital between February 1974 and August 1987 and selected, for a more detailed review, those in whom cryotherapy had been performed. Each patient had undergone a complete ocular examination under anesthesia according to a protocol that we have previously published.^{1,2} Fundus drawings depicting the size and location of each tumor were made, and the size of each tumor was determined by indirect ophthalmoscopy. Ultrasonography was also used in the larger tumors. The tumor diameters were carefully estimated in disk diameters and converted to millimeters, based on the assumption that one disk diameter was equal to about 1.5 mm. Tumor thickness was carefully estimated to the nearest 0.5 mm based on the clinical experience of one of us (J.A.S.). Most tumors treated with cryotherapy were too small and peripheral to obtain accurate clinical photographs and ultrasonographic thickness measurements, but ultrasonography was routinely attempted for all tumors that were estimated to be greater than 2 mm thick.

Cryotherapy was administered by the triple freeze-thaw technique using nitrous oxide as described in the literature.² With the patient under general anesthesia, indirect ophthalmoscopy was used. The cryoprobe was used as a scleral depressor, and the tumor was elevated on the tip of the cryoprobe (Figs. 1 and 2). The cryoprobe was applied until the tumor and the

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